Adam, gradients and parameter updates: full task, all layers

c1 | c2 | f1 | f2 | head

Relative weight update ‖ΔW‖ / ‖W‖

Bias update RMS (includes B)

Mean Adam weight denominator

RMS(gradient / Adam denominator)

Weight-gradient RMS

0 | 10k | 20k | 30k

Update

CH′ self | CH′ ← CH′+B response | CH′+B self | CH′+B ← CH′ response

Every75-update epoch is summarized without smoothing. Thin lines: every seed; thick: all10 mean. Y scales are shared per metric row, symlog(linthresh1e−10).
Δb includes the subsequent B projection for CH′+B recipients. The Adam denominator is sqrt(v/(1−β₂ᵗ))+ε. These are diagnostics, not isolated causal effects.